

VVD Data Dictionary & Schema Reference

Purpose

This document maps the VVD (Virtual Visa Debit) vintage engine's success logic to the Success Library Table Data Assets framework. It serves as the formal data dictionary for all VVD data assets - source tables, success metrics, campaign mappings, and output schemas - providing the team a single reference to understand how VVD success is detected, calculated, and reported.

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1. Architecture Overview

The VVD vintage engine implements the SuperFact four-layer architecture in code:

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| LAYER 1: EXPERIMENT METADATA |
| Source: DTZTA_T_TACTIC_EVNT_HIST (HIVE) |
| Question: "Who is in the test?" |
| Output: Client-level experiment assignments with group & cohort |
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| LAYER 2: CAMPAIGN MAPPING |
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| Source: CAMPAIGN_METADATA (hardcoded in engine) |
| Question: "What do we measure?" |
| Output: Mnemonic -> primary/secondary metric mapping |
|-----|
| LAYER 3: SUCCESS DEFINITIONS (Logic Repo) |
| Source: SUCCESS_DEFINITIONS (hardcoded in engine) |
| Question: "How do we calculate success?" |
| Output: CLNT_NO + SUCCESS_DT per metric |
|-----|
| LAYER 4: CLIENT JOURNEY |
| Source: VENDOR_FEEDBACK_MASTER/EVENT (EDW) |
| Question: "What did clients do?" |
| Output: Engagement flags (sent, opened, clicked, unsubscribed) |
|-----|
| ANALYSIS LAYER: VINTAGE ENGINE |
| Joins: Layer 1 + Layer 3 + Layer 4 |
| Processing: Cumulative success rate by day |
| Output: Vintage curves + channel breakdown |
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Current state: All mappings and definitions are hardcoded in the vintage engine Python code. This document extracts and formalizes them for the transition to the governed Success Library.

2. Campaign-to-Metric Mapping

C. campaign_mapping (VVD)

Maps each VVD mnemonic to its success metrics. This is the VVD-specific implementation of the whiteboard's Table C.

mnemonic	campaign_name	success_type	1st_success (primary)	2nd_success (secondary)	metric_code_primary	metric_code_secondary
VCN	VVD Contextual Notification	ACQUISITION	card_acquisition	--	VVD_ACQ_001	--
VDA	VVD Black Friday Cyber Monday T	ACQUISITION	card_acquisition	--	VVD_ACQ_001	--
VDT	VVD Activation Trigger	ACTIVATION	card_activation	--	VVD_ACT_001	--
VUI	VVD Usage Trigger	USAGE	card_usage	wallet_provisioning	VVD_USG_001	VVD_PRV_001
VUT	VVD Tokenization Usage Campaign	TOKENIZATION	wallet_provisioning	card_usage	VVD_PRV_001	VVD_USG_001
VAW	VVD Add To Wallet Contextual Not	TOKENIZATION	wallet_provisioning	card_usage	VVD_PRV_001	VVD_USG_001

Metric Reuse Matrix

Shows which metrics are shared across campaigns:

Metric	VCN	VDA	VDT	VUI	VUT	VAW
card_acquisition (VVD_ACQ_001)	P	P				
card_activation (VVD_ACT_001)			P			
card_usage (VVD_USG_001)				P	S	S
wallet_provisioning (VVD_PRV_001)				S	P	P

P = Primary, S = Secondary

3. Source Data Assets

3.1 HIVE Sources (Parquet on HDFS)

DTZTA_T_TACTIC_EVNT_HIST - Experiment Population

Attribute	Value
Purpose	Identifies which clients are in each experiment
Layer	1 (Experiment Metadata)
Path	/prod/sz/tsz/00150/cc/DTZTA_T_TACTIC_EVNT_HIST/
Partition	EVNT_STRT_DT=YYYY*
Grain	One row per client per tactic event

Column	Type	Description	Used As
TACTIC_ID	string	Full tactic identifier (mnemonic at pos 8-10)	Campaign identifier
TACTIC_EVNT_ID	string	Client identifier (requires trim + leading zero st	Derives CLNT_NO
TREATMT_STRT_DT	date	Treatment start date	Success window start
TREATMT_END_DT	date	Treatment end date	Success window end
TST_GRP_CD	string	Test group code (TG4 = test)	Action/Control split
RPT_GRP_CD	string	Report group code	Cell-level detail
TACTIC_CELL_CD	string	Tactic cell code (contains channel info)	Channel identification
EVNT_STRT_DT	date	Event start date	Partition key

Derived columns:

Derived Column	Transformation	Description
CLNT_NO	regex_replace(trim(TACTIC_EVNT_ID), '^0+', '')	Client number
MNE	substring(TACTIC_ID, 8, 3)	Campaign mnemonic
COHORT	date_format(TREATMT_STRT_DT, 'yyyy-MM')	Monthly cohort

WINDOW_DAYS	datediff(TREATMT_END_DT, TREATMT_STRT_DT)	Treatment window length
CHANNEL	Extracted from TACTIC_CELL_CD	Channel (email = "EM")

DDWTA_VISA_DR_CRD - Visa Debit Card Data

Attribute	Value
Purpose	Card issuance and activation events
Layer	3 (Success Definitions)
Path	/prod/sz/tsz/00050/data/DDWTA_VISA_DR_CRD/Partitio
Partition	CAPTR_DT=YYYY*
Grain	One row per card
Used by	VVD_ACQ_001 (card_acquisition), VVD_ACT_001 (card_

Column	Type	Description	Used As
CLNT_NO	string	Client number	Join key
ISS_DT	date	Card issue date	SUCCESS_DT for acquisition
ACTV_DT	date	Card activation date	SUCCESS_DT for activation
STS_CD	string	Card status code	Filter: '06' (Active), '08' (Approved)
SRVC_ID	int	Service identifier	Filter: 36 (Visa Direct / VVD)
CAPTR_DT	date	Capture date	Partition key

DDWTA_T_PT_OF_SALE_TXN - Point of Sale Transactions

Attribute	Value
Purpose	Card usage / transaction events
Layer	3 (Success Definitions)
Path	/prod/sz/tsz/00050/data/DDWTA_T_PT_OF_SALE_TXN/SNA
Partition	SNAP_DT=YYYY*
Grain	One row per transaction
Used by	VVD_USG_001 (card_usage)

Column	Type	Description	Used As
CLNT_CRD_NO	string	Client card number (contains CLNT_NO at pos 7-15)	Derives CLNT_NO
TXN_DT	date	Transaction date	SUCCESS_DT
TXN_TP	int	Transaction type	Filter: 10, 13, 12
MSG_TP	string	Message type	Filter: '0210', '0220'
AMT1	decimal	Transaction amount	Filter: > 0
SRVC_CD	int	Service code	Filter: 36 (Visa Direct)
SNAP_DT	date	Snapshot date	Partition key

Client extraction: CLNT_NO = CAST(SUBSTR(CLNT_CRD_NO, 7, 9) AS INTEGER) with leading zeros stripped.

3.2 EDW Sources (Teradata)

DDWV05.CLNT_CRD_POS_LOG - Card POS Log

Attribute		Value	
Purpose		Wallet provisioning events (zero-amount token tran	
Layer		3 (Success Definitions)	
Schema		DDWV05	
Grain		One row per POS transaction	
Used by		VVD_PRV_001 (wallet_provisioning)	

Column	Type	Description	Used As
CLNT_CRD_NO	string	Client card number	Derives CLNT_NO; filter prefix '45190'
VISA_DR_CRD_NO	string	Visa debit card number	Filter prefix '45199'
TOKN_REQSTR_ID	string	Token requestor ID	Join to TOKEN_LIST; filter first char > '0'
TXN_DT	date	Transaction date	SUCCESS_DT
AMT1	decimal	Transaction amount	Filter: = 0 (provisioning event)
POS_ENTR_MODE_CD_NON_EMV	string	POS entry mode	Filter: '000' (token provisioning)
SRVC_CD	int	Service code	Filter: 36 (Visa Direct)

DL_DECMAN.TOKEN_LIST - Token Registry

Attribute		Value	
Purpose		Token/wallet provisioning registry	
Layer		3 (Success Definitions)	
Schema		DL_DECMAN	
Grain		One row per token	
Used by		VVD_PRV_001 (wallet_provisioning) - joined to CLNT	

Column	Type	Description	Used As
TOKEN_ID	string	Token identifier	Join key to TOKN_REQSTR_ID
TOKEN_WALLET_IND	string	Wallet provisioning indicator	Filter: 'Y'

DTZV01.VENDOR_FEEDBACK_MASTER / VENDOR_FEEDBACK_EVENT - Email Engagement

Attribute	Value
Purpose	Email engagement tracking (sent, opened, clicked,
Layer	4 (Client Journey)
Schema	DTZV01
Grain	One row per client per disposition event
Used by	Engagement curves in vintage engine

Column	Type	Description	Used As
CLNT_NO	string	Client number	Join key
TREATMENT_ID	string	Treatment/tactic identifier	Campaign association
disposition_cd	int	Disposition type code	See disposition codes below
disposition_dt_tm	datetime	Event timestamp	Engagement date

Disposition codes:

Code	Meaning	Derived Flag	Derived Date Field
1	Sent	EMAIL_SENT	EMAIL_SENT_DT
2	Opened	EMAIL_OPENED	EMAIL_OPENED_DT
3	Clicked	EMAIL_CLICKED	EMAIL_CLICKED_DT
4	Unsubscribed	EMAIL_UNSUBSCRIBED	EMAIL_UNSUBSCRIBED_DT

4. Success Metric Definitions

Each metric follows a standard contract: given a date range, return CLNT_NO + SUCCESS_DT for all clients who met the success criteria.

VVD_ACQ_001 - Card Acquisition

Attribute	Value
Metric ID	VVD_ACQ_001
Metric Name	Card Acquisition
Pillar	Conversion
Business Definition	Client acquired a new VVD card (issued in active/a
Success Date	ISS_DT (card issue date)
Source	HIVE - DDWTA_VISA_DR_CRD
Grain	Client (first issue date per client)
Campaigns	VCN (primary), VDA (primary)

Filters:

Field	Operator	Value	Reason
STS_CD	IN	('06', '08')	Card must be Active or Approved
SRVC_ID	=	36	Visa Direct / VVD product only
ISS_DT	IS NOT NULL	--	Card must have been issued

Output contract: CLNT_NO (string), SUCCESS_DT (date)

VVD_ACT_001 - Card Activation

Attribute	Value
Metric ID	VVD_ACT_001
Metric Name	Card Activation
Pillar	Conversion
Business Definition	Client activated their VVD card (first use after i
Success Date	ACTV_DT (card activation date)
Source	HIVE - DDWTA_VISA_DR_CRD
Grain	Client (first activation date per client)
Campaigns	VDT (primary)

Filters:

Field	Operator	Value	Reason
STS_CD	IN	('06', '08')	Card must be Active or Approved
SRVC_ID	=	36	Visa Direct / VVD product only
ISS_DT	IS NOT NULL	--	Card must have been issued first

Note: Same source table and filters as VVD_ACQ_001 but uses ACTV_DT instead of ISS_DT. Activation is a downstream event from acquisition.

Output contract: CLNT_NO (string), SUCCESS_DT (date)

VVD_USG_001 - Card Usage

Attribute	Value
Metric ID	VVD_USG_001
Metric Name	Card Usage
Pillar	Engagement
Business Definition	Client used their VVD card for a point-of-sale tra

Success Date	TXN_DT (transaction date)
Source	HIVE - DDWTA_T_PT_OF_SALE_TXN
Grain	Client (first transaction date per client)
Campaigns	VUI (primary), VUT (secondary), VAW (secondary)

Filters:

Field	Operator	Value	Reason
SRVC_CD	=	36	Visa Direct / VVD product only
AMT1	>	0	Must be a real transaction, not zero-amount
TXN_TP + MSG_TP	IN	See below	Valid purchase transaction types

Valid transaction types:

TXN_TP	MSG_TP	Description
10	0210	Standard purchase
13	0210	E-commerce purchase
12	0220	Recurring / subscription

Client extraction: CLNT_NO = regexp_replace(SUBSTR(CLNT_CRD_NO, 7, 9), '^0+', '')

Output contract: CLNT_NO (string), SUCCESS_DT (date)

VVD_PRV_001 - Wallet Provisioning

Attribute	Value
Metric ID	VVD_PRV_001
Metric Name	Wallet Provisioning
Pillar	Conversion
Business Definition	Client provisioned their VVD card to a digital wal
Success Date	TXN_DT (provisioning transaction date)
Source	EDW - DDWV05.CLNT_CRD_POS_LOG JOIN DL_DECMAN.TOKEN
Grain	Client (first provisioning date per client)
Campaigns	VUT (primary), VAW (primary), VUI (secondary)

Filters:

Field	Operator	Value	Reason
AMT1	=	0	Provisioning events are zero-amount
SUBSTR(CLNT_CRD_NO, 1, 5)	=	'45190'	VVD card BIN prefix
SUBSTR(VISA_DR_CRD_NO, 1, 5)	=	'45199'	VVD debit card BIN verification
SUBSTR(TOKN_REQSTR_ID, 1, 1)	>	'0'	Valid token requestor

POS_ENTR_MODE_CD_NON_EMV	=	'000'	Token provisioning POS mode
SRVC_CD	=	36	Visa Direct / VVD product only
TOKEN_WALLET_IND	=	'Y'	Confirmed wallet provisioning

Join: CLNT_CRD_POS_LOG.TOKN_REQSTR_ID = TOKEN_LIST.TOKEN_ID

Client extraction: CLNT_NO = CAST(SUBSTR(CLNT_CRD_NO, 7, 9) AS INTEGER)

Why EDW instead of HIVE: Token registry (DL_DECMAN.TOKEN_LIST) is only available in EDW. The join between POS log and token list is necessary to confirm wallet provisioning via TOKEN_WALLET_IND.

Output contract: CLNT_NO (string), SUCCESS_DT (date)

5. Experiment & Population Schema

A. experiment_mapping (VVD)

Populated from DTZTA_T_TACTIC_EVNT_HIST with mnemonic filter.

Column	Type	Source	Description
experiment_id	string	Derived	{MNE}_{COHORT} (e.g., VCN_2025-06)
mnemonic	string	SUBSTRING(TACTIC_ID, 8, 3)	Campaign mnemonic (VCN, VDA, VDT, VUI, VUT, VAW)
test_type	string	--	Not currently captured in vintage engine
hypothesis	string	--	Not currently captured in vintage engine
lift_type	string	--	Not currently captured in vintage engine
measurement	string	Implicit	Test vs. no contact (TG4 vs others)
sub_type	string	--	Not currently captured in vintage engine

Gap: The vintage engine does not store experiment design metadata (hypothesis, test_type, lift_type). These fields exist in the whiteboard schema but are not populated by the engine. They would need to come from tech specs or an intake form.

B. experiment_population (VVD)

One row per client per experiment assignment.

Column	Type	Source	Description
experiment_id	string	Derived	Links to experiment_mapping

clnt_no	string	regex_replace(trim(TACTIC_EVNT_ID), '^0+', '')	Client number
tactic_id	string	TACTIC_ID	Full tactic identifier
start_dt	date	TREATMT_STRT_DT	Treatment start date
end_dt	date	TREATMT_END_DT	Treatment end date
group	string	TST_GRP_CD	Test group (TG4 = Action, others = Control)
rpt_grp_cd	string	RPT_GRP_CD	Report group / cell detail
cohort	string	date_format(TREATMT_STRT_DT, 'yyyy-MM')	Monthly cohort
window_days	int	datediff(TREATMT_END_DT, TREATMT_STRT_DT)	Treatment window in days
channel	string	Extracted from TACTIC_CELL_CD	Channel (e.g., email = "EM")

6. Output Schemas

6.1 Vintage Curves (Primary Output)

The main output of the vintage engine. One row per mnemonic/cohort/group/metric/day combination.

Column	Type	Description	Example
MNE	string	Campaign mnemonic	VCN
COHORT	string	Treatment month (yyyy-MM)	2025-06
TST_GRP_CD	string	Test group code	TG4
RPT_GRP_CD	string	Report group code	RPT01
METRIC	string	Metric type	PRIMARY, SECONDARY, EMAIL_SENT, EMAIL_OPEN, EMAIL_CLICK, EMAIL_UNSUB
DAY	int	Days since treatment start	0, 1, 2, ..., 90
WINDOW_DAYS	int	Total treatment window (median)	90
CLIENT_CNT	int	Total clients in this cell	15000
SUCCESS_CNT	int	Cumulative successes by this day	450
RATE	decimal(8,4)	Success rate percentage	3.0000

Metric types explained:

METRIC value	Source	Denominator	Numerator
PRIMARY	Campaign's 1st_success metric	All clients in cell	Clients with success event
SECONDARY	Campaign's 2nd_success metric	All clients in cell	Clients with success event
EMAIL_SENT	Engagement data	All clients targeted with email	Clients who received email
EMAIL_OPEN	Engagement data	Clients who received email	Clients who opened
EMAIL_CLICK	Engagement data	Clients who received email	Clients who clicked
EMAIL_UNSUB	Engagement data	Clients who received email	Clients who unsubscribed

6.2 Channel Breakdown

One row per mnemonic/cohort/group/channel combination.

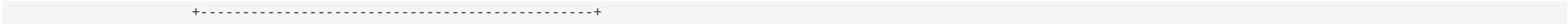
Column	Type	Description	Example
MNE	string	Campaign mnemonic	VCN
COHORT	string	Treatment month	2025-06
TST_GRP_CD	string	Test group code	TG4
RPT_GRP_CD	string	Report group code	RPT01
CHANNEL	string	Channel from TACTIC_CELL_CD	EM
CLIENT_CNT	int	Clients in channel	8000
SUCCESS_CNT	int	Successes in channel	240
RATE	decimal(6,2)	Success rate percentage	3.00

6.3 Client-Level Success (Intermediate)

Intermediate dataset created during processing. One row per client per metric.

Column	Type	Description
CLNT_NO	string	Client number
MNE	string	Campaign mnemonic
COHORT	string	Treatment month
TST_GRP_CD	string	Test group code
RPT_GRP_CD	string	Report group code
WINDOW_DAYS	int	Treatment window
SUCCESS_FLAG	int	1 if success occurred, 0 otherwise
FIRST_SUCCESS_DT	date	Earliest success date within window
DAYS_TO_FIRST_SUCCESS	int	Days from treatment start to first success
SUCCESS_COUNT	int	Number of success events within window
EMAIL_SENT	int	1 if email was sent
EMAIL_OPENED	int	1 if email was opened
EMAIL_CLICKED	int	1 if email was clicked
EMAIL_UNSUBSCRIBED	int	1 if client unsubscribed

7. Entity Relationship Map



```

|          CAMPAIGN_METADATA (Layer 2)          |
| VCN -> card_acquisition (P)                   |
| VDA -> card_acquisition (P)                   |
| VDT -> card_activation (P)                   |
| VUI -> card_usage (P), wallet_provisioning (S)|
| VUT -> wallet_provisioning (P), card_usage (S)|
| VAW -> wallet_provisioning (P), card_usage (S)|
|-----+-----+
|          | selects metric                      |
|          v                                    |
+-----+-----+
| TACTIC_EVNT_HIST | | SUCCESS_DEFINITIONS (Layer 3) |
| (Layer 1 - HIVE) | |                               |
|                  | | VVD_ACQ_001 ? DDWTA_VISA_DR_CRD |
| CLNT_NO          | | VVD_ACT_001 ? DDWTA_VISA_DR_CRD |
| TACTIC_ID        | | VVD_USG_001 ? DDWTA_T_PT_OF_SALE |
| TREATMT_STRT_DT  | | VVD_PRV_001 ? CLNT_CRD_POS_LOG   |
| TREATMT_END_DT   | |          + TOKEN_LIST           |
| TST_GRP_CD       | |                               |
| RPT_GRP_CD       | | Output: CLNT_NO + SUCCESS_DT     |
|-----+-----+
|          |                                     |
|          +-----+                             |
|          | JOIN CONDITION |                   |
|----->| ON CLNT_NO |<-----+
|          | WHERE SUCCESS_DT |                 |
|          | BETWEEN          |                 |
|          | TREATMT_STRT_DT |                 |
|          | AND              |                 |
|          | TREATMT_END_DT  |                 |
|          +-----+
|          |
|          v
+-----+
| CLIENT-LEVEL SUCCESS |
| (Intermediate Dataset) |
|                       |
| CLNT_NO, MNE, COHORT, |
| TST_GRP_CD, RPT_GRP_CD, |
| SUCCESS_FLAG,          |
| FIRST_SUCCESS_DT,      |
| DAYS_TO_FIRST_SUCCESS  |
|-----+-----+

```

			+-----+	+-----+	+-----+	
			v	v	v	
+-----+		+-----+		+-----+		
VINTAGE		CHANNEL		ENGAGEMENT		
CURVES		BREAKDOWN		CURVES		
DAY-level		Per-channel		EMAIL_SENT		
cumulative		aggregation		EMAIL_OPEN		
rates				EMAIL_CLICK		
-----+		-----+		EMAIL_UNSUB		
				-----+		

8. Business Rules Reference

8.1 Global Constants

Constant	Value	Description
YEARS_TO_INCLUDE	[2025, 2026]	Data years loaded from HIVE partitions
AGGREGATION_LEVEL	MONTH	Cohorts are monthly (yyyy-MM)
TEST_GROUP_CODE	TG4	TG4 = test/action group; all others = control
VVD_SERVICE_CODE	36	Visa Direct / VVD product identifier
CARD_BIN_PREFIX	45190	VVD client card BIN
DEBIT_BIN_PREFIX	45199	VVD Visa debit card BIN

8.2 Success Window Rule

All success metrics follow the same temporal window rule:

SUCCESS is counted IF AND ONLY IF:
SUCCESS_DT >= TREATMT_STRT_DT
AND SUCCESS_DT <= TREATMT_END_DT

First success date per client is used for vintage curve day calculation:

DAYS_TO_FIRST_SUCCESS = datediff(FIRST_SUCCESS_DT, TREATMT_STRT_DT)

8.3 Client Number Extraction Rules

Source Table	Extraction Method
TACTIC_EVNT_HIST	regexp_replace(trim(TACTIC_EVNT_ID), '^0+', '')
DDWTA_VISA_DR_CRD	Direct CLNT_NO column
DDWTA_T_PT_OF_SALE_TXN	regexp_replace(SUBSTR(CLNT_CRD_NO, 7, 9), '^0+', '')
CLNT_CRD_POS_LOG	CAST(SUBSTR(CLNT_CRD_NO, 7, 9) AS INTEGER)

8.4 Engagement Denominator Rules

Metric	Denominator	Numerator
EMAIL_SENT	All clients targeted with email channel	Clients with disposition_cd = 1
EMAIL_OPEN	Clients with EMAIL_SENT = 1	Clients with disposition_cd = 2
EMAIL_CLICK	Clients with EMAIL_SENT = 1	Clients with disposition_cd = 3
EMAIL_UNSUB	Clients with EMAIL_SENT = 1	Clients with disposition_cd = 4

9. Open Questions

From Whiteboard (Applicable to VVD)

#	Question	Status	Impact
1	Should 1st_success, 2nd_success in tactic_history	Open	Affects how View 3 and View 4 aggregate
2	How does the VVD vintage output map to the tactic_	Open	Vintage curves are day-level; tactic_history is cl

From Vintage Engine Implementation

#	Question	Status	Impact
3	Experiment metadata (hypothesis, test_type, lift_t	Open	Layer 1 schema is incomplete without it
4	The engine hardcodes TST_GRP_CD = TG4 as test. Is	Open	Affects Action/Control classification
5	VVD_USG_001 transaction types: TXN_TP=13 is labele	Open	Affects what counts as "usage"
6	Wallet provisioning uses AMT1 = 0 to identify prov	Open	Could create false positives
7	The engine currently produces CSV output. What is	Open	Affects the delivery/output schema

Gaps Between Vintage Engine and Target Architecture

#	Gap	Current State	Target State
1	Campaign mapping	Hardcoded Python dict	campaign_mapping table (Table C)
2	Success definitions	Hardcoded Python dict	GitHub Logic Repo (Layer 3)
3	Experiment metadata	Not captured	experiment_mapping table (Table A)
4	Cross-sell view	Not implemented	cross_sell crosstab pivot (Table 5)
5	Share of Wallet	Not implemented	SOW Programs + SOW Usage tables
6	Channel Interactions	Partial (email only)	Full channel tracking table
7	Daily automated refresh	Manual notebook execution	Scheduled pipeline

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